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Development of an optimally designed real-time automatic citrus fruit grading–sorting machine leveraging computer vision-based adaptive deep learning model

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ABSTRACT

Conventional automation approaches for postharvest operations are plagued by time and data inefficiency seldom leading to suboptimal solutions. Automatic machines often require highly skilled software professionals for calibration and reconfiguration thus making the technology prone to high costs.

Contemporary sensors and smart devices capable of handling deep learning image analytics have been employed in the present study for the development of an automatic machine that performs postharvest operations, like—washing, vision-based sorting and weight-based grading of citrus fruits with much reduced human effort while achieving excellent performance for the designated tasks. Accuracy of performance was ensured by the optimal design of mechanical components carried out by kinematic synthesis and dimensional analysis.

The machine was equipped with an effective custom lightweight CNN model "SortNet" that was designed and tuned to carry out vision-based classification of citrus fruits into "accept" and "reject" based on surface characteristics. SortNet was less complex and took less computational time while exhibiting comparable accuracy with respect to existing state-of-the-art pre-trained deep learning models. An embedded system operated by a single-board computer was used in the weight grading section for segregating fruits based on three weight categories. Evaluation, realization and transferability of the above said strategy was demonstrated by the real hardware with physical actuators working in real-time to serve as proof-of-concept for a sustainable solution to postharvest automation of citrus fruits.

1. Introduction

Fruits and vegetables remain physiologically and metabolically active even after harvest and can be exposed to numerous abiotic stresses during handling, storage and transportation (Hodges, 2003). These stresses can result in accelerated metabolic responses, degradation of organic and inorganic substrates, and an increase in the rate of decay (Tanou et al., 2017); often leading to nutritional (energy, micro-, macro- and phyto-nutrients) and quality (physical, biochemical and flavour attributes) degradation as well as economic loss. This can be a reason for the fact that India's global export of fruits and vegetables is an abysmal 2 per cent, despite having a respectable share of 15 and 11 per cent as compared with global fruits and vegetable production, respectively (Bhardwaj, 2021). In the developing world including India, on-farm washing, grading and sorting of fruits is carried out by unskilled agricultural workers. Poor handling as well as the subjectivity of manual grading paralyzes the quality of material, decreasing its appeal and market potential. There is also an element of drudgery involved in these tasks as well. Also, staring at us is a massive depletion in the agricultural workforce which was about 60 per cent of the total workforce at the start of this century and shall be halved to 30 per cent by 2050 (Mehta et al., 2014). In this context, it deems appropriate for different actors like researchers and innovators to incorporate newer techniques and technologies for the betterment of the farming community in particular and the health of the agricultural economy of the developing world as a whole (Cofre-Bravo et al., 2019).

Automation is the mantra across the entire agricultural mechanization environment. The use of sensors and embedded systems has never been as popular as it is today. In a postharvest vision-based sorting and weight grading system, input devices, - camera and load cell with associated image processing algorithm, embedded system, actuators, all connected to an optimally designed mechanical system can form

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a potent combination for an "Automatic Fruit Grader (AFG)". This machine shall prove to be a potential solution to accelerate the process for uniformity in product packaging and automatic fruit classification in an on-farm postharvest processing environment.

Computer vision systems have been demonstrated to be effective for quality inspection of various postharvest operations (Patrício and Rieder, 2018; Tripathi and Maktedar, 2020). These systems can replicate the human vision system by using standard approaches and consequently, eradicating the labourious human inspection tasks which are limited by varied senses of faculties amongst individuals. The machine learning/deep learning driven by the image dataset is a new area for fruit classification. This method is less expensive because it requires only a low-cost digital camera and a controller (Sustika et al., 2018).

Recently, artificial intelligence (AI) approaches represented by deep learning models have achieved significant results in the field of food quality inspection through advanced computer vision algorithms (Fan et al., 2020; Fu et al., 2022; Minh Trieu and Thinh, 2021). As far as vision-based intelligent systems are concerned, convolutional neural network (CNN) models are well recognized in the agricultural research community as a potential approach for image classification (Paymode and Malode, 2022). CNN provides automatic feature extraction and requires very little pre-processing. The input data is initially forwarded to the feature extraction section and extracted features are forwarded into the classifier network (Navamani, 2019). The automatic learning feature of deep learning models makes it a very suitable approach in the area of postharvest quality management of agricultural produce, in terms of identification and morphological features (Zhou et al., 2019; Rong et al., 2020). Deep learning solutions shall facilitate the vision-based application with better accuracy and sturdiness for postharvest applications. Different AI-based machine learning (ML) models for automatically grading fruits based on their physical and chemical features have been developed in many recent studies (Aguilar et al., 2021; Arboleda et al., 2021; Chopra et al., 2021; Siricharoen et al., 2023).

ML techniques can approximate experimental results when a basic model is created and can be tuned using conventional optimization algorithms. However, traditional algorithms have some drawbacks, such as difficulty in navigating error function plateaus and getting stuck in local extremum. These algorithms can be modified with meta-heuristic ones to address these problems (Katebi et al., 2020). ML techniques have been used more frequently for various applications in the prediction, identification and classification field in recent years (Armaghani et al., 2020; Mohammadhassani et al., 2013; Milovančević et al., 2019; Safa and Kachitvichyanukul, 2019; Shariati et al., 2019, 2020a,b,c, 2021a; Suhatril et al., 2019; Toghroli et al., 2014).

CNN is a subset of DL techniques; it contains several non-linear transformation functions. Due to the depth and weight sharing between nodes within the network, CNNs can automatically perform both feature extraction and selection, which can reduce over-fitting (Nasiri et al., 2019). Due to the above features, Deep CNN (DCNN) architectures can be efficiently applied for classification, prediction and object detection in numerous scientific, agricultural, food processing, biomedical and engineering applications (Tang et al., 2023; Chakraborty et al., 2022; Madani et al., 2022; Wang et al., 2022; Chen et al., 2021; Kunkel et al., 2021; Madani et al., 2021; Shariati et al., 2021b; Safa et al., 2020, 2016; Mohammadhassani et al., 2014). There are different DCNN pre-trained models AlexNet, GoogLeNet, ResNet50, VGG16 and 19, which can also be used for similar applications (Nasiri et al., 2019; Raikar et al., 2020). The DL-based AI models can overcome the complex feature extraction activity of ML. These models are selected for their exceptional automated feature extraction abilities, simple and effective training of the raw images using the best computing resources, and transferability to edge computing devices.

Washing agricultural produce not only improves aesthetics but also ensures hygienic and safe food (Maffei et al., 2016; Buchholz et al., 2011). Fruit colour is an indicator of ripeness (Chu et al., 2022), juiciness (Wu et al., 2019), oil content (Septiarini et al., 2021) and stress (Hamdani et al., 2021); fruit weight is normally considered a criterion for purchase, sale and packaging (Boyette et al., 1996). A newly developed farm-savvy automatic machine, "Automatic Fruit Grader (AFG)" is reported in this study; wherein washing, vision-based sorting and weight grading all the operations can be carried out real-time in a single go. Washing will ensure the removal of adhering foreign material while diminishing the field heat as well, vision-based sorting shall help in classifying the fruits as "accept" and "reject" depending on user-defined criteria, weight grading shall segregate the "accept" into three user-defined weight categories. Vision-based sorting will be accomplished by the development of a custom light-weight DCNN model; such models have high efficiency and low latency, and can be easily deployed on an edge device like raspberry pi (Li et al., 2022; Razfar et al., 2022). The developed machine will be unique in its functionality as it includes both vision-based and weight-based grading in a single unit. Evaluation of the machine will be reported in terms of different parametric performance factors.

2. Materials and methods

2.1. Embedded system for colour sorting and weight grading

The embedded system comprised three subsystems: image acquisition system (IAS), weight measurement system (WMS) and fruit segregation system (FSS). Singulated fruit coming from the washer in two channels would pass through IAS, WMS and FSS before finally falling through designated chutes. The fruits were sorted and graded based on their surface feature and weight, respectively. Image-based segregation classified "accept" (Fig. S1) as the citrus fruits which had consistent orange intact blemish-free peel; otherwise, they were considered as "reject" (Fig. S1). The weight-based grading grouped (Annon, 2010) citrus fruits into three weight (g) categories, large(L) $\geq$ 140, 140 > medium(M) $\geq$ 100 and small(S) < 100. The flow of information within the embedded system has been explained in Fig. 1.

2.1.1. Image acquisition system (IAS)

Acquisition of images was carried out in an enclosed inspection compartment with a camera (Ravtron, webcam RAV17/02, 1MP, KSG Automation Pvt. Ltd., New Delhi, India) and LED (5 V, 1.2 W e-Cosmos Solutions, Bengaluru, India) light. Illumination inside the enclosure was 40 lux (LX-103, HTC Instruments, Mumbai, India). The camera and LED were mounted on the ceiling of the IAS enclosure for each channel (Fig. 2a). During the capture of images, fruits in two channels were rolling while moving in a forward direction towards the weighing pan custom-designed conveyor with optimized rotational and transitional speed. The channels were partitioned by an opaque object; each had a camera and a light source of its own. The single-board-computer (SBC) Raspberry pi 4B (64 bit, 2 GB RAM, Quad core Cortex-A72 ARM v8, Raspberry pi Foundation, Cambridge) working on the Raspbian operating system activated the camera, recorded the 10 frames which were resized to 125 $\times$ 125 and processed it using a deep learning algorithm. The SBC controller retrieved the result produced by the image classification subprogram and worked in sync with the weight measurement system.

The fruits moved beneath the camera in the forward direction while rotating on multiple axes on a roller conveyor driven by a chain and sprocket assembly powered by a 2.24 kW electric motor through a variable frequency drive (Emotorn VS, Comptron Greaves, Bengaluru, India). The ends of the rollers of the conveyor had raised collar of 15 mm height (Fig. S2). One of the collars was toothed to act as a pinion for the rack underneath. This rack and pinion system enabled the rollers to rotate while moving in the forward direction (Fig. 2b). Effective vision-based sorting by the machine relied on feeding the algorithm with a variety of images of the same fruit. In other words, a fruit should rotate an optimum number of times while it travels through the field of vision (FOV) of the camera. Designed experiments (Table 1) were

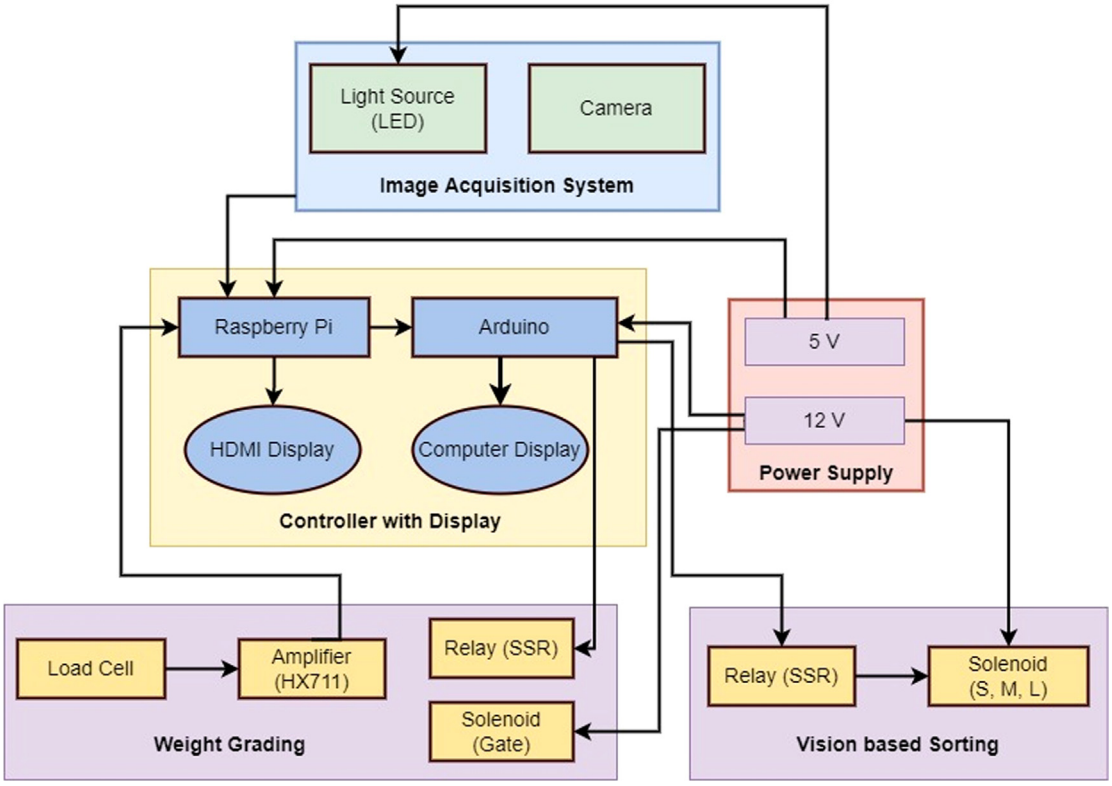


Fig. 1. Block diagram of embedded system for weight grading and vision based sorting.

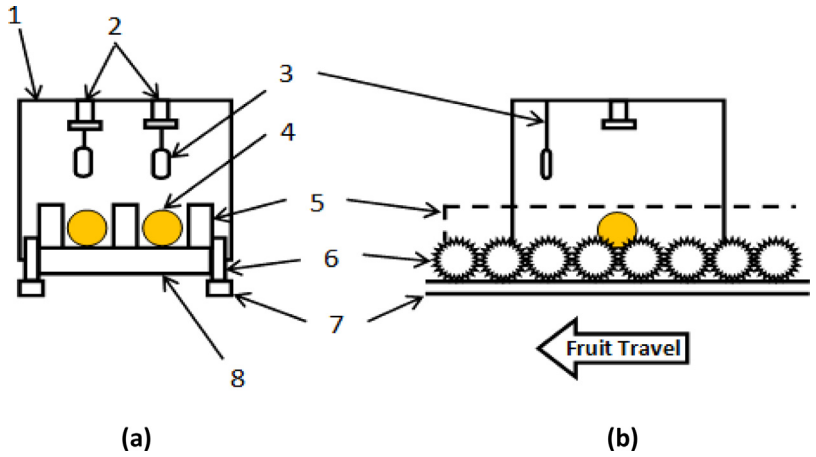


Fig. 2. Schematic representation of the image acquisition system (a) Front view (b) Side view. 1. Image acquisition enclosure, 2. Camera, 3. LED light, 4. Fruit, 5. Opaque partition, 6. Collar of roller with tooth (pinion), 7. Rack, 8. Roller.

conducted with varying, - roller conveyor speed (by changing input electric current frequency with a variable frequency drive), number of rolls of rollers per unit length (by using rollers with different numbers of pinions) and FOV (by changing the height of the camera above conveyor). Response surface methodology (Box and Wilson, 1951) was used to optimize these parameters with respect to the accuracy and capacity of the machine.

2.1.2. Weight measurement system (WMS)

After the image of the fruit has been acquired, fruit leaves the roller conveyor (Fig. 3a) to fall on the weighing pan while it is stopped by a "gate" (Fig. 3b). The weight of fruits in the two channels was measured by cantilever strain sensor type load cells (SunRobotics, 10 kg, 1.0 ± 0.1 mV/V, single point). The weight of the fruit was recorded as an average of 10 readings. The output from the load cell was connected to an amplifier (HX711, Sparkfun, 24-bit) module (Pestana et al., 2016) before the signals were passed on to SBC. The General-Purpose Input/Output (GPIO) pins (GPIO14, GPIO15, GPIO17 and GPIO23) of SBC are directly connected with digital pins (3, 4, 5 and 6) of Arduino (Mega 2560, 16 MHz, 256 kb flash memory, 8 kb SRAM, 8 bit, Arduino, Ivera, Italy). As soon as SBC provides a high output to GPIO pins, the corresponding Arduino pin is actuated to finally excite the solid-state relay (SSR 40DD, Input: 3–32 V DC, Output: 12–32 V DC, 40 ampere, Fotek, Taipei, Taiwan) to actuate a solenoid (12 V DC, 10 amperes, 7 kg, Kalyan Engineers, Pune, India) operating the "gate" and "plunger" for any weight more than 40 g (Fig. 3c) and the fruit subsequently moves on to belt conveyor (Fig. 3d) and enters the FSS. This would be followed by actuating any one (out of three) solenoid of both channels to push the fruit off the belt conveyor in a designated (S, M and L) chute depending on the weight category of that particular

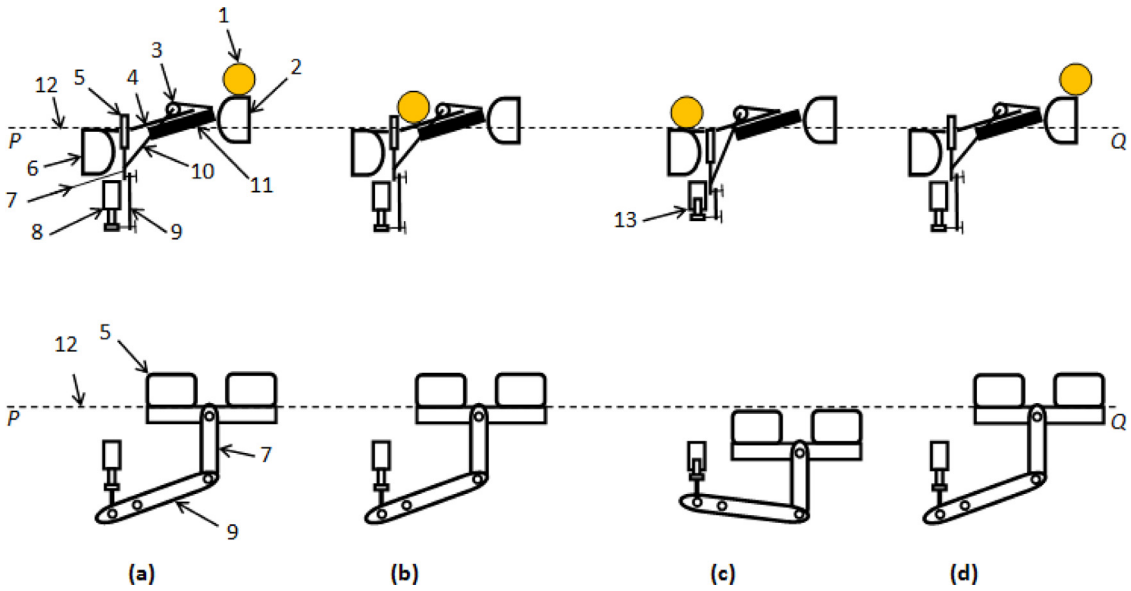


Fig. 3. Schematic representation of fruit movement, plunger action and "gate" mechanism (a) Gate closed, fruit on roller conveyor (b) Gate closed, fruit on weighing pan (c) Gate open, fruit pushed away from weighing pan by plunger (d) Gate closed, next fruit on roller conveyor. 1. Fruit, 2. Roller conveyor, 3. Plunger, 4. Weighing pan, 5. Gate, 6. Belt conveyor, 7. Gate link, 8. Solenoid, 9. Solenoid link, 10. Plunger link, 11. Load cell, 12. Reference plane (PQ), 13. Solenoid in ON position with "Gate" pulled down and "Plunger" pushed forward.

Table 1
Experimental design for optimized performance of vision-based sorting system of the machine.

Code for the values	Machine parameters		
	Belt speed, m/min	FOV, mm	Roller revolution, No./m
−1	2	120	3.8
0	4	140	4.5
+1	6	160	5.2
No. of experiments	Coded machine parameters		
	X_1	X_2	X_3
14	±1	±1	±1
6	0	0	0
20	**Total number of experiments**		

Code '0' is for the central point of the parameter range being investigated, ±1 for factorial points, X_1: Belt speed, X_2: Field of vision (FOV) and X_3: Roller revolution.

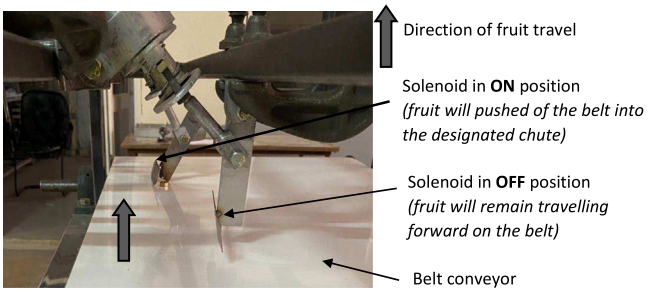


Fig. 4. Position of the solenoid for segregating the fruits in the designated chutes as per the weight category on the belt conveyor.

fruit. The code for weight measurement was developed in Python 3.7 and the code for solenoid actuation for fruit segregation on a weight basis (Fig. 4) was done in Arduino 1.8.13.

Before use, load cells were calibrated with standard weight (200 g) placed as load, initialized calibration factor by 1 and then executed calibration program of HX711 library. The calibration factor obtained by digital value (return from calibration program) and known weight

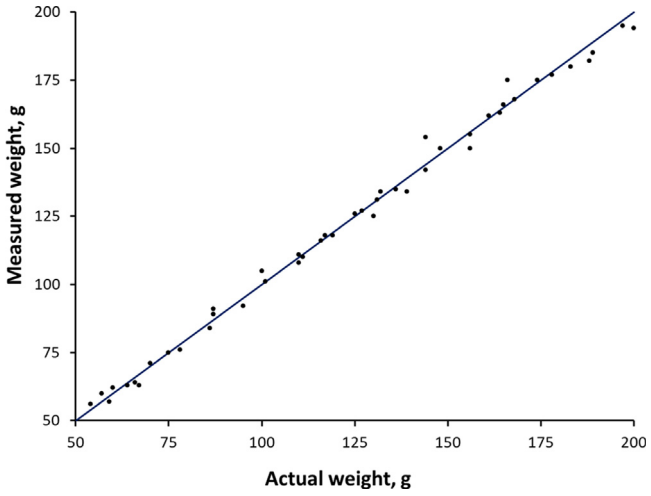


Fig. 5. Comparison plot of actual versus measured weight.

can be expressed as, -

$$C = \frac{x}{y} \tag{1}$$

where, digital value (x) = 130 000, standard weight (y) = 200 g and calibration value (C) = 650. The x value obtained from the calibration program was read by the controller as input from the load cell. This final calibration factor value was used by the HX711 library to convert voltage into weight value. The accuracy of WMS was estimated by comparing the weight with the actual weight (Fig. 5). The error (50 samples) was estimated by considering the difference between the measured value (results obtained by the controller) and the actual value (standard weight).

The synchronized action of the "gate" and "plunger", effecting the required vertical displacement of the "gate" (which was more than that offered by the solenoid) and lateral movement of the "plunger" (to offer just the right displacement to the fruit so that it reaches the belt conveyor) required right dimensions and mechanisms. Kinematic synthesis (Russell et al., 2013) was used for identifying the dimensions

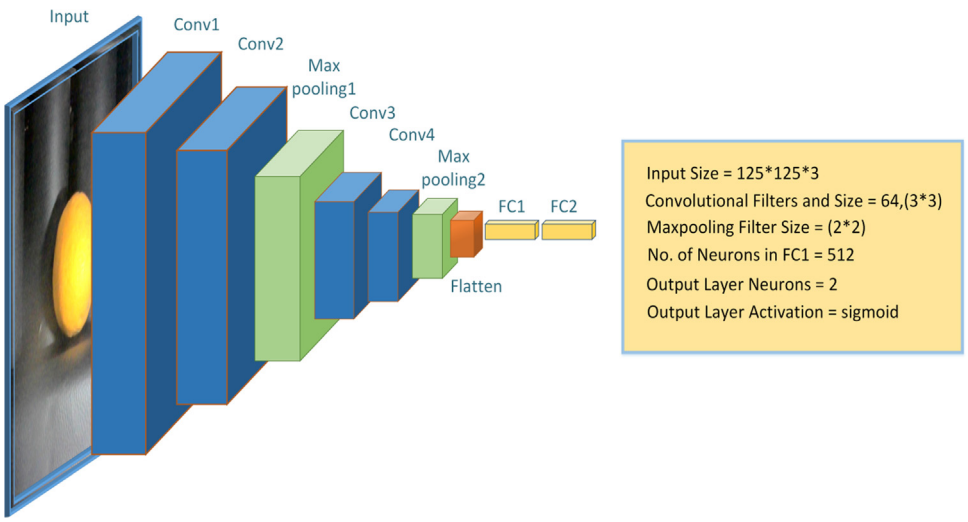


Fig. 6. SortNet – Lightweight custom CNN architecture for fruit sorting.

of this four-bar mechanism which included path, motion and function generation. Precision in the working of said mechanism was rendered by dimensional synthesis (Hernández et al., 2021), which took care of the link dimensions, positions, angles and precision points. All the synthesis and analysis were carried out in MATLAB R2020b (Matlab, 2020).

2.2. Image processing in a deep learning environment

2.2.1. Development of custom CNN model

The key objectives of developing a simple custom-lightweight CNN model over a pre-trained or sophisticated model were to improve the performance over a custom orange dataset and reduce overall latency with a minimum process time. Several custom CNN architectures were designed to evolve a fresh CNN model for the vision sorting of citrus fruits. This was carried out after tuning parameters, like—the number of CNN layers, activation function, filter size, dropout rate, epochs and batch size. The developed '**SortNet**' model is capable of classifying citrus fruits into 'accept' and 'reject' classes with high accuracy and low response time.

The architectural overview of SortNet is shown in Fig. 6 and the detailed architecture of the proposed model along with it is a fresh configuration, - including the layers, filters, output shape and number of parameters are shown in Fig. 7. The input size used was 125×125 to reduce processing time and provide faster responses to micro-controller for timely actuation. The architecture consists of 4 convolutional layers and 2 max pooling layers. All convolutional operations in the architecture are followed by a ReLU (Rectified Linear Unit) activation function. The sigmoid function was used as an activation function in the output layer. All convolutional layers used a 3×3 filter with a stride of 1, and include 1-pixel padding. The number of filters used in the convolutional layer was 64 and the max-pooling filter size was set as 2×2. A dropout layer was added for avoiding overfitting and the model includes 27,672,002 trainable parameters. The experiments were conducted on a desktop computer (Intel Core i7 Processor with base frequency 2.60 GHz, 16 GB RAM, 6 GB Nvidia GeForce GTX 1660 Ti GPU) with the operating system Windows 10 (64 bits). The training and performance evaluation of the developed CNN model was conducted in Python 3.7 with libraries Tensorflow 2.2, Keras 2.3.1, Opencv 3.4, etc.

2.2.2. Model performance

The performance of the CNN model for classification tasks is generally evaluated using a confusion matrix. The test dataset was used for classifying the images into new images. In this study, it was a binary classification problem and each image in the test set of the sample was assigned to one of the classes, i.e., either 'accept' or 'reject'. The indices used to define the performance of the model are,

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{2}$$

$$\text{Precision} = \frac{TP}{TP + FP} \tag{3}$$

$$\text{Sensitivity/Recall} = \frac{TP}{TP + FN} \tag{4}$$

$$\text{F1 Score} = \frac{2 * \text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

where, TP is True Positive, indicating the number of samples correctly predicted by the model as "accept"; TN denotes True Negative indicating the number of correct predictions by the model, as "reject". FP is False Positive denotes the number of misclassifications of citrus fruits which belong to the "accept" category but are classified as "reject". FN or False Negative was the number of misclassified citrus fruits which belong to the "reject" category but were classified as "accept".

2.3. Working principle

Although the machine (AFG) discussed in this paper can handle any agricultural commodity with a specific gravity less than unity and sphericity more than 0.7; experiments were conducted with 100 kg citrus (*Citrus reticulata*) fruits from the orchard of a local farmer at Khamkheda (23°23′15.4″N 77°26′29.4″E), Bhopal, India. Fruits straight from the orchard can be put in a continuous recirculating washing tank (Fig. 8, 7) of 500 L working capacity with 10 per cent circulation (Mundi et al., 2017); wherein water washing of fruits was accomplished by six pressure jets capable of removing all the dirt adhering to the surface of fruits and driving the floating bed of fruits towards singulation conveyor (Fig. 8, 9). The clean fruits are picked up by a moving inclined two-row singulation conveyor belt comprising cups (60 mm, diameter and 15 mm, depth) lined along the periphery of the belt and separated by a longitudinal distance of 300 mm (Chakraborty et al., 2017). The fruits are finally dropped onto the roller conveyor (Fig. 8, 10) whereon the fruits rotate as they travel forward into the image acquisition enclosure (Fig. 8, 5). The fruit conveyors were made of food-grade belts (endless PVC cleated, thickness of 3 mm and width of 30 mm) and the roller conveyor was made of food-grade nylon. The material of construction of the machine components and parts was stainless steel (SS 304). The rotational speed of the roller was optimized with respect to frame capture rate to ensure an error free vision-based sorting along with high capacity. The fruits kept moving ahead in two channels before being stationed by the closed gate at the weighing pan (Fig. 8, 4). After the weight of the fruit

Layer (type)	Output Shape	Param #
conv2d_1 (Conv2D)	(None, 125, 125, 64)	1792
activation_1 (Activation)	(None, 125, 125, 64)	0
conv2d_2 (Conv2D)	(None, 123, 123, 64)	36928
activation_2 (Activation)	(None, 123, 123, 64)	0
max_pooling2d_1 (MaxPooling2	(None, 61, 61, 64)	0
dropout_1 (Dropout)	(None, 61, 61, 64)	0
conv2d_3 (Conv2D)	(None, 61, 61, 64)	36928
activation_3 (Activation)	(None, 61, 61, 64)	0
conv2d_4 (Conv2D)	(None, 59, 59, 64)	36928
activation_4 (Activation)	(None, 59, 59, 64)	0
max_pooling2d_2 (MaxPooling2	(None, 29, 29, 64)	0
dropout_2 (Dropout)	(None, 29, 29, 64)	0
flatten_1 (Flatten)	(None, 53824)	0
dense_1 (Dense)	(None, 512)	27558400
activation_5 (Activation)	(None, 512)	0
dropout_3 (Dropout)	(None, 512)	0
dense_2 (Dense)	(None, 2)	1026
activation_6 (Activation)	(None, 2)	0

Total params: 27,672,002
Trainable params: 27,672,002
Non-trainable params: 0

Fig. 7. Architecture of the DCNN model (SortNet) used for image classification.

has been measured in both channels, the gate opens and the fruit is pushed the to belt conveyor (Fig. 8, 1), where solenoids (Fig. 8, 16) are actuated for "accept" and based on weight category of fruits, push them into designated chutes (Fig. 8, 2); "reject" would travel straight to the chute (Fig. 8, 19) meant for the same. The entire sequence of operation of the machine along with the flow of information and decision-making hierarchy has been explained in Fig. 9.

2.4. Man-machine interfacing

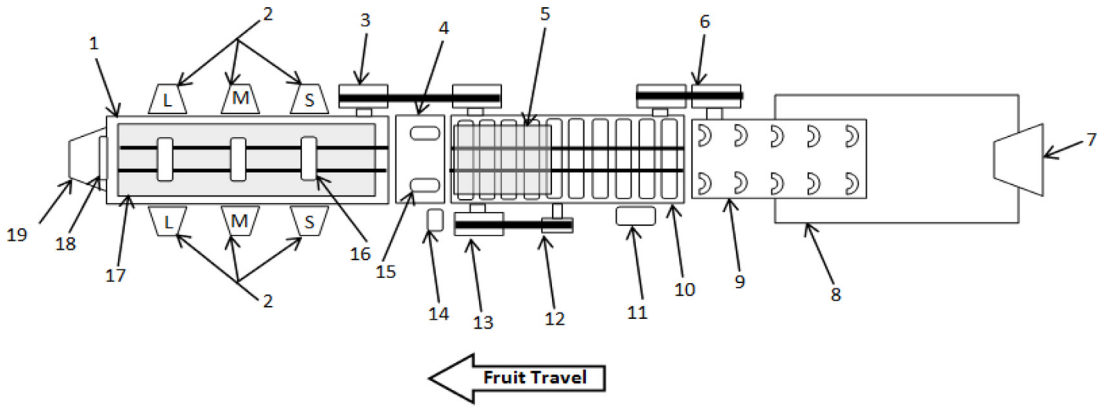
2.4.1. Operating the machine

A graphical user interface (GUI) was developed for easy operation of the machine, which included access to choose the fruit in an HDMI display (7 inch, capacitive touch, 1024 × 600, Waveshare Shenzhen, China). Hyper Text Markup Language (HTML), Cascading Style Sheets (CSS) and Javascript were used to develop the GUI. It offers the user access to backend code for making changes to fruit-specific choices, in terms of weight categories and criteria for "accept" and "reject". The layout of the web application was created using HTML scripts; while the dynamic functionality was derived from integrated JavaScript code. The bootstrap framework was employed for the rapid designing of a web application to make it responsive, consistent and free of repetitions. The home page of the GUI contains a start menu to initialize the sorting program (Fig. 10a), next page provides options for the selection of fruit type (Fig. 10b), followed by a screen with prediction results. The user-selected input is sent to the backend Python software, where the fruit-specific model was activated and results are saved before being sent to the HTML page for display.

2.4.2. Assessment of ergonomics

It was hypothesized that the developed machine, "Automatic Fruit Grader" which carries out washing, vision-based sorting and weight grading (three categories) in one single go will make the whole sequence of operation faster and shall minimize the drudgery involved with these tasks when they are done manually. An ergonomic study was conducted in terms of physiological cost or workload and fatigue or comfort offered by the machine (*if any*) in comparison to the same set of tasks (washing, vision-based sorting and weight grading) carried out manually. Also, the correctness of the identification of "accept" and "reject" and judgment-based grading of fruits into three weight categories by human subjects were compared with that of the machine.

A total of twelve physically and medically fit male subjects aged 18–45 years and BMI 18–23.2 kg/m^2 were selected for ergonomic



(a)



(b)

Fig. 8. (a) Line diagram of the machine explaining the various parts and the mode of action, (b) The machine, "Automatic Fruit Grader" in actual form. 1. Belt conveyor; 2. "Accept" chutes for three weight categories (L, Large; M, Medium; S, Small); 3. Drive pulley for the belt conveyor; 4. Weight grading assembly; 5. Image acquiring console; 6. Drive for the singulation conveyor; 7. Fruit feeding chute; 8. Washing tank; 9. Fruit singulation conveyor; 10. Roller conveyor; 11. Control panel; 12. Power source (AC 3 phase, 3 hp electric motor); 13. Pulley drive for the roller conveyor; 14. Gate solenoid actuators; 15. Load cells; 16. Fruit deflection solenoid actuators; 17. Housing for the electronic circuits and DC power source; 18. Touch screen; 19. Discharge chute for the "Reject".

assessment of tasks (washing, vision-based sorting, and weight grading) with 100 kg citrus fruits that would otherwise be carried out by the machine in one go. The VO_2max (L/min), the upper limit of oxygen consumption was determined for the selected subjects in laboratory conditions following the Naughton exercise protocol on a computerized treadmill (LE 200CE Jaeger, Hochberg, Germany). The oxygen consumed by the subject and corresponding heart rate during the exercise on the treadmill was measured using a metabolic measurement system (K4b^2 Mobile Breath, Breath Metabolic System, COSMED Pulmonary Function Equipment, Italy) and polar heart rate monitor (Polar RS 400, Polar Electro Oy, Kempele, Finland).

At the onset, subjects were briefed about the entire methodology of the task to be carried out. Meanwhile, resting heart rate, working heart rate, time required for the task were recorded. Work pulse i.e., the increase in heart rate over resting heart rate (ΔHR, beats/min) during the experiments was calculated by subtracting the resting heart rate from the working heart rate of the subjects (Gite et al., 2020). The energy equivalent of one-litre oxygen, i.e., 20.9 kJ was used for the calculation of the energy consumption rate (ECR, kJ/min) expended during the task. It was calculated using the following equation (Gite et al., 2020).

$$ECR = VO_2 \times 20.9 \tag{6}$$

Work output has been expressed in terms of per cent of maximum aerobic capacity ($\%VO_2max$) of subjects (Potdar et al., 2021) as, -

$$\%VO_2max = \frac{VO_2}{VO_2max} \times 100 \tag{7}$$

The cardiac cost (beats/100 kg fruits) was used for comparing the drudgery involved in two methods of operation (manual versus machine) and was calculated by, -

$$Cardiac\ cost = \frac{\Delta HR \times 3600}{Work\ output} \tag{8}$$

Workload evaluation techniques include classification of work severity based on published guidelines for heart rate, oxygen consumption rate and percentage of VO_2 max. The acceptable workload for Indian workers is consuming 35% of VO_2 max while they work (Gite et al., 2020). The muscle fatigue during the task was measured by surface electromyography (sEMG) using a 16-channel wireless system comprising a universal serial bus dongle (DG2, DataLITE Pioneer, Biometrics Ltd, Newport, UK) attached to a PC along with battery operated

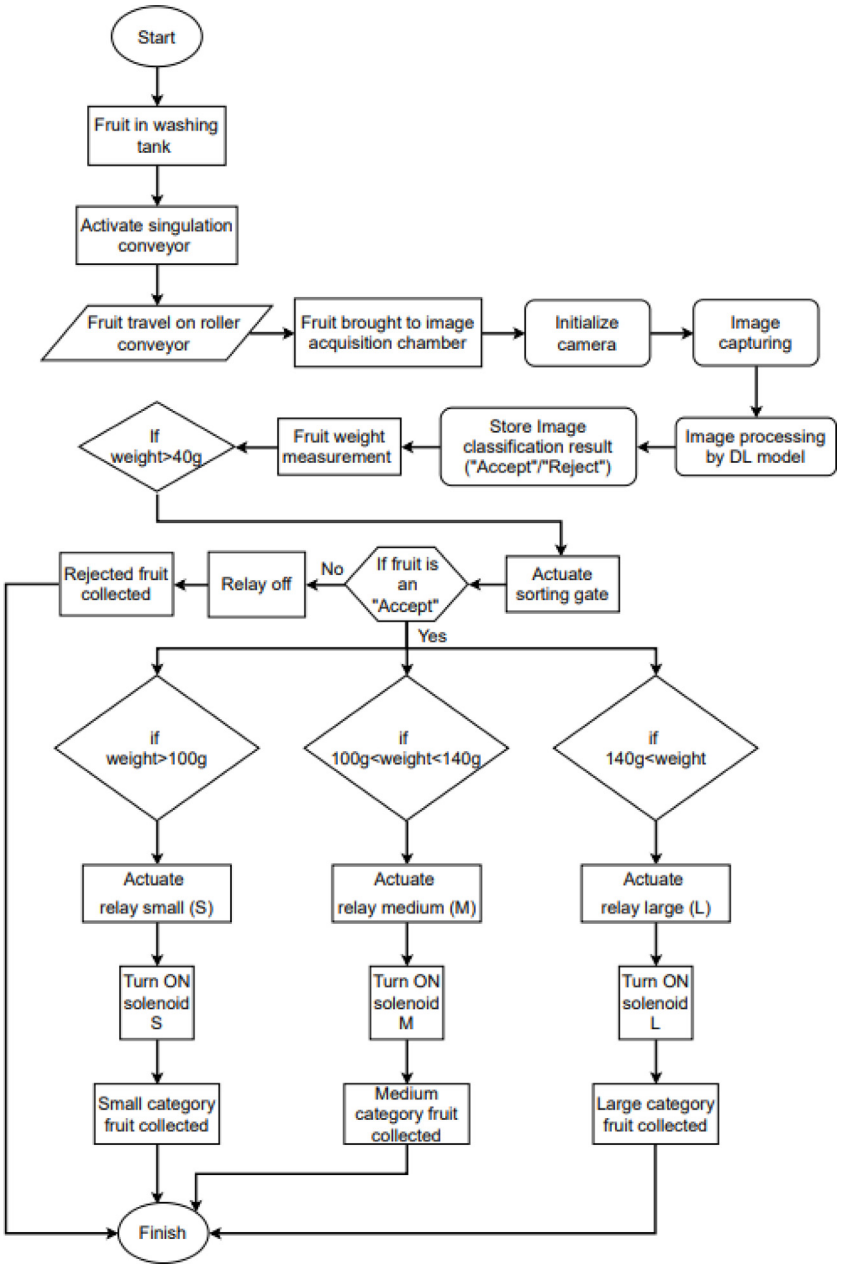


Fig. 9. Flow chart of machine operation sequence and decision making for vision-based fruit sorting and weight measurement.

wireless sEMG sensors (10 to 490 Hz, DataLITE LE230-Biometrics Ltd, Newport, UK), and sampled at a rate of 2000 samples per second per channel. Four sEMG sensors were placed on the skin of subject at extensor carpi ulnaris (right), extensor radialis ulnaris (right), extensor carpi ulnaris (left) and extensor radialis ulnaris (left) muscles under investigation. Prior to placing sEMG sensor, the skin was cleaned with alcohol to remove dead skin, oils and dirt from electrode sites. Before start of experiments, the subjects performed maximum voluntary contraction (MVC) to obtain reference voluntary electrical activity (RVE) of selected muscles. The response of muscle under investigation was quantified in per cent change of MVC while the task was performed traditionally and while with the developed machine.

3. Results and discussion

The results of the machine were evaluated in terms of its mechanical functionality, accuracy for vision-based sorting, grading into three weight categories, software systems, reduction in human labour and time-saving.

3.1. Machine performance

The developed "Automatic Fruit Grader" underwent a series of tests with citrus fruits in batches of 100 kg and working for more than 50 h. This ensured robustness of the machine and assembly of the embedded system (Fig. 11) for on-farm activity.

3.1.1. Vision-based sorting

The key element of vision-based system was the placement of camera and capture of image under uniform lighting. In this machine, camera was mounted firmly on the ceiling of image acquisition chamber fixed on screws with adequate padding to attenuate vibrations of machine. Flexible dark curtains were placed at the entry and exit of travelling fruits to keep image capture independent of ambience light conditions. The success of the algorithm to extract the feature from

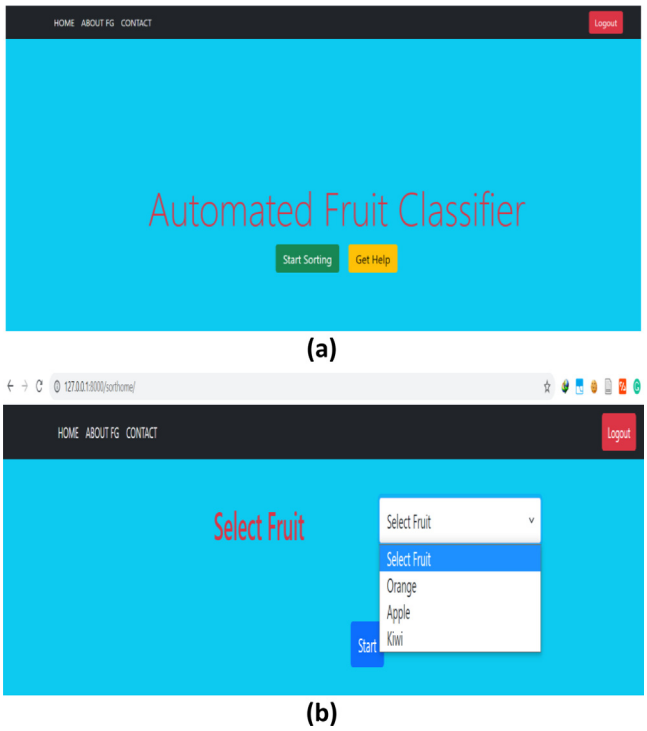


Fig. 10. Graphical user interface for operating the machine (a) Home screen of the touch screen (b) Fruit selection screen.

images and accuracy of vision-based system to rightly classify the fruits will be heavily dependent on the amount of surface area of fruit that is being captured by the camera.

Speed of roller conveyor (belt speed, m/min), placement of camera above the conveyor will decide the field of vision (FOV, mm) and number of rolls the fruit will undergo underneath the camera will depend upon the number of times the rollers roll (revolutions per minute). These three factors will have a critical role in making vision-based system accurate. Response surface methodology (RSM) was used to study the effect of belt speed, FOV and roller revolution on accuracy and capacity (kg/h) of the machine. A quadratic model was fitted to experimental data and ANOVA was done for statistical significance of the model (Table 2). The developed models of accuracy and capacity were statistically significant ($p < 0.01$). The lack of fit was insignificant indicating that developed models were correct and able to predict responses. The coefficient of determination (R^2) of both the models was >99% and coefficient of variation (cv) was less than 2.5%, indicating that the model is adequate enough to optimize variables and predict responses.

Roller conveyor belt speed was an important parameter that directly affected capacity as well as accuracy of the automatic fruit grader. ANOVA shows that the effects of belt speed ($p < 0.01$), FOV ($p < 0.01$) and roller revolution ($p < 0.05$) had a significant effect on accuracy of machine. The accuracy of vision-based sorting system increased to 77 per cent with an increase in belt speed till 4 m/min, beyond this speed inaccurate vision-based sorting results were obtained. The FOV of camera was found to have an effect on accuracy. As shown in Fig. 12a, accuracy is linearly correlated with FOV. The lowest accuracy was found at low values of FOV, whereas, it was highest (96 per cent) at high (160 mm) values of FOV. The exposure of the fruit was greater at longer FOV lengths, so the camera collected a greater number of frames; thus an increased accuracy of vision-based sorting system. The roller revolution of conveyor system is another important parameter that affected the accuracy. An increase (5.2 No. of rolls per m) in roller revolutions increased (81 per cent) accuracy of the

Table 2
Regression analysis of second order polynomial models in terms of accuracy and capacity.

Predictor	β (Coded factor)	
	Accuracy	Capacity
Intercept	77.78**	225.01**
X_1	2.84**	91.61**
X_2	20.06**	0.6
X_3	1.48*	0.71
X_1X_2	0.19	−0.062
X_1X_3	1.66*	0.71
X_2X_3	1.44*	0.79
X_1^2	−8.65**	1.83
X_2^2	−2.45*	1.28
X_3^2	1.55	−1.27
ANOVA		
Model (F-value)	157**	1526**
Coefficient of determination (R^2), %	99.3	99.9
Coefficient of variation (cv), %	2.47	1.09

X_1: Belt speed, X_2: FOV, X_3: Roller revolution.
*significant at $p < 0.05$.
**significant at $p < 0.01$.

Table 3
Confusion matrix for vision-based sorting of citrus fruits.

Actual			
	Accuracy (%) = 97.0	Accepts (200 fruits)	Rejects (100 fruits)
Predicted	Accepts	195[0.98]	3
	Rejects	5	97[0.95]

Values in the square brackets indicate the "reliability".

grader. Increasing revolution of the roller led to more rotation of the fruit during conveyance under the camera FOV. So, frames of different orientations of citrus fruit were obtained and an increased accuracy was observed. Fig. 12a shows that all the factors affected accuracy but FOV was most predominant factor responsible for variations in the accuracy of the system. The interaction effect of roller revolution with belt speed ($p < 0.05$) and FOV ($p < 0.05$) was found to have a significant effect on the machine capacity. Higher revolution of roller at medium belt speed gave better accuracy. On the other hand, an increase in roller revolution with an increased FOV resulted in better accuracy of vision-based system.

The effect of belt speed on machine capacity was significant but FOV and roller revolution had no significant effect on machine capacity. It has been seen from Fig. 12b that capacity of the machine was found to be linearly correlated with belt velocity. It increased from 135.1 kg/h to 321.4 kg/h by increasing belt speed from 2 to 6 m/min. The optimum values of independent variables were obtained with the help of numerical and graphical optimization techniques of RSM. The optimum values of belt velocity, FOV and roller revolution were obtained as 4.0 m/min, 160 mm and 4.95 no./m, respectively. At these optimal values of the variables, predicted values of accuracy and capacity were obtained as 97.7 per cent and 228.7 kg/h, respectively.

The machine was tested with 300 citrus fruits in five runs under the obtained optimum conditions to ascertain accuracy for vision-based sorting into the two categories, "accept" and "reject". Sample size comprised 200 fruits of "accept" and 100 fruits of "reject" category. The results of experiments were observed as "reliability" and "accuracy" of classification by the machine, while former defines ratio of the correctly predicted category to the sum of predictions of same category; the latter explains ratio of the sum total of correctly predicted to that of the total subjects. It was observed that "reliability" of the machine for classification of fruits as "accept" and "reject" was 0.98 and 0.95, respectively; accuracy of the machine for vision-based sorting was as high as 97.0 per cent (Table 3).

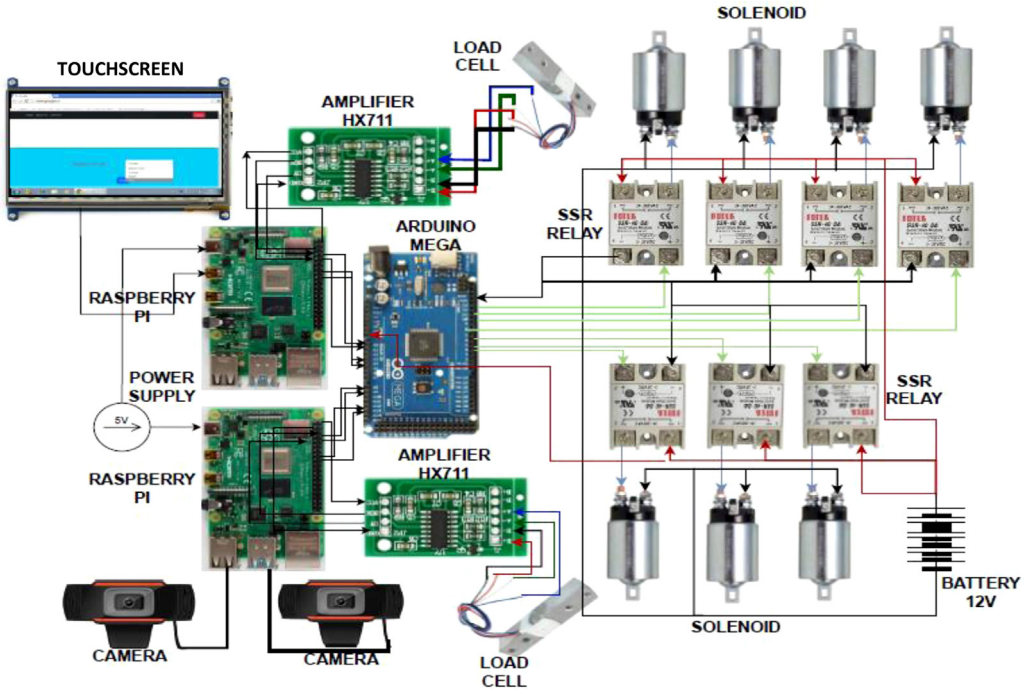


Fig. 11. Representative circuit diagram of the electronic components, sensors and other gadgets used in "Automatic Fruit Grader".

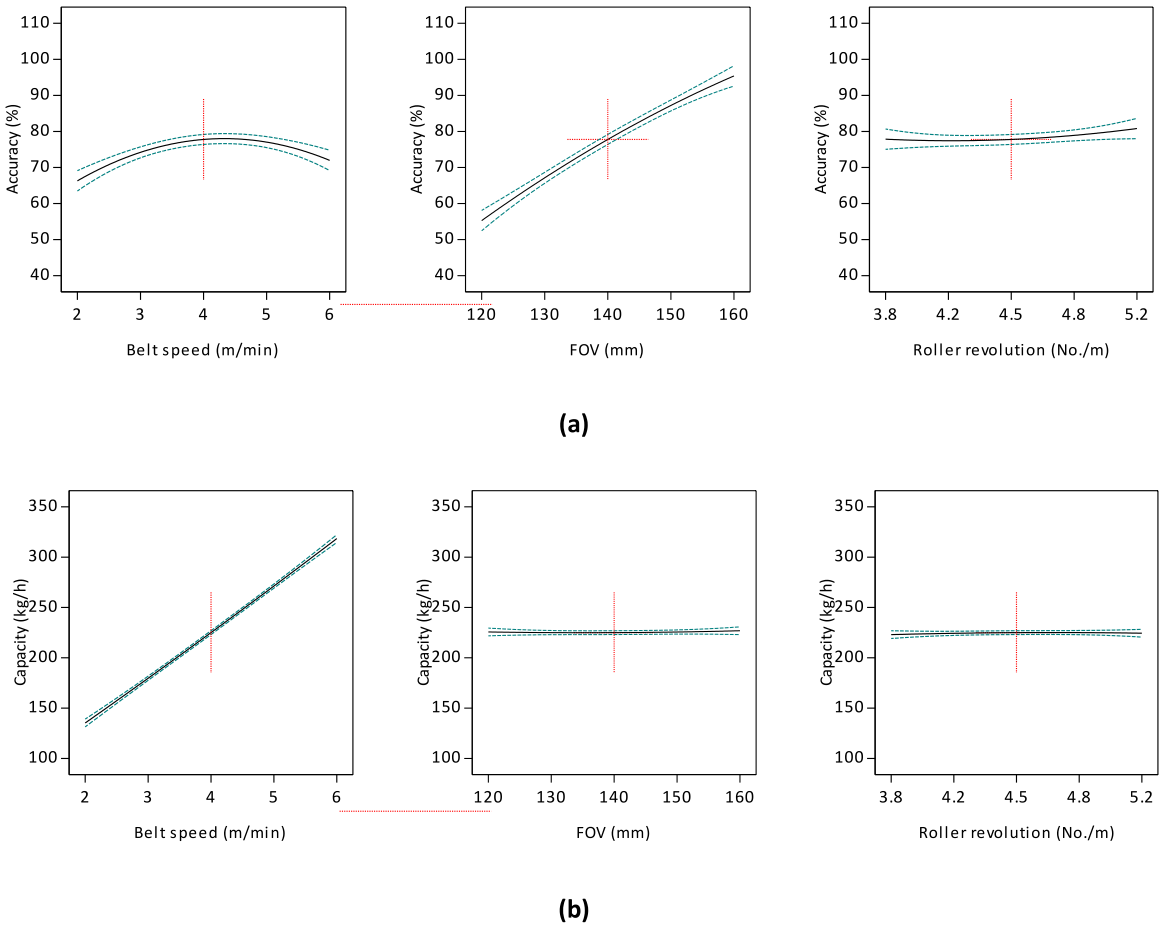


Fig. 12. Effect of different parameters on the machine (a) Accuracy (b) Capacity.

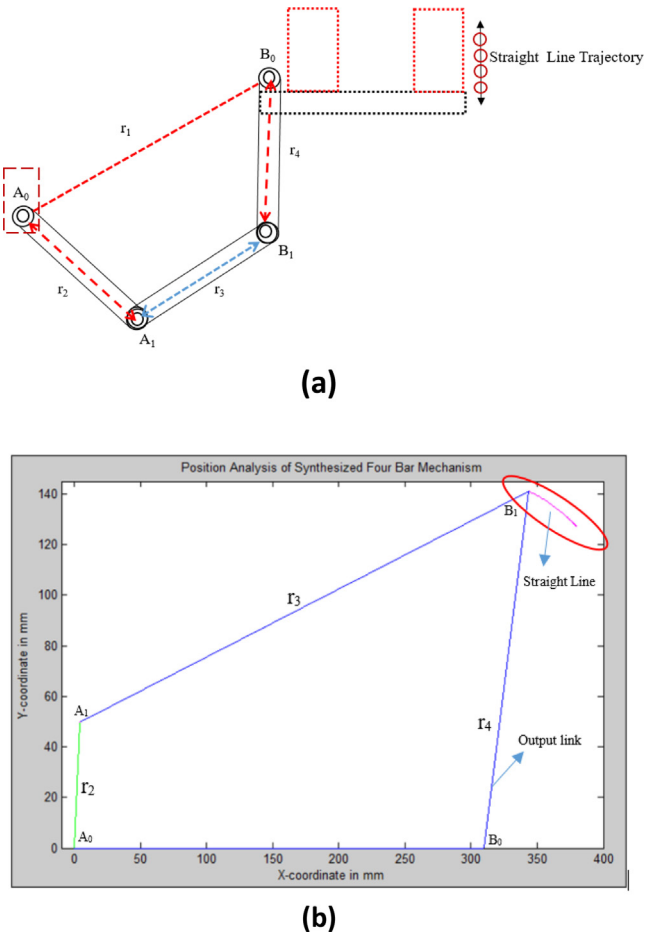


Fig. 13. Extraction of dimensions of the linkages of gate and plunger in the weight grading system (a) Straight line trajectory of prescribed four bar mechanism (b) Synthesized four bar mechanism.

3.1.2. Weight grading

Fruits would slide down the roller conveyor onto weighing pan where weight of fruit will be recorded as an average of 10 values. Load cells were firmly attached to the body of machine, rubber washers were used at fastening points to check incidence of vibrations affecting the correctness of measure. Post measurement of weight, the plunger would push the fruit forward with simultaneous pulling down of gate, all in a single motion being effected by a solenoid actuated by SBC being signalled by load cell. The links responsible for pulling the gate and thrusting the plunger forward were considered as a four-bar mechanism of a closed linkage system (Pickard et al., 2020). Output from mechanism was obtained through coupler and rocker end whichever was suitable as per the requirement of operation. This planar linkage was synthesized by three methods such as graphical, analytical, and optimal methods. Kinematic synthesis was performed by selecting prescribed precision points of straight-line trajectory as it was for the gate (Fig. 13a). Three precision points were selected for synthesis of the mechanism and was animated in MATLAB R2020b (Matlab, 2020) to verify compatibility of the mechanism with respect to the precision points. Based on these precision points on the trajectory, a proper analytical synthesis procedure was performed and the dimensions of the mechanism were obtained. It was derived that the gate link (Fig. 3a, 7) was 145 mm, solenoid link (Fig. 3a, 9) was 351 mm and the position of fulcrum on it was 50 mm from solenoid (Fig. 13b).

The machine was tested with 300 citrus fruits in five runs to assess the accuracy for weight grading into the three prescribed categories (S, M and L). Sample size comprised 100 fruits of each category. The results of experiments were observed as "reliability" and "accuracy" of classification by the machine. It was observed that "reliability" of machine for S, M and L was 0.96, 0.87 and 0.92, respectively. The accuracy of machine for categorization of citrus fruits into three weight categories was 91.3 per cent (Table 4).

3.2. CNN model performance

A total of 1280 images comprised the dataset for gauging the performance of CNN model, 80 per cent of data are used for training (1024) the model; 10 per cent for validation (128) and 10 per cent for testing (128). While training the model, images were manually labelled into two classes "accept" and "reject" (Fig. S1). The model was trained for 150 epochs with a batch size of 32 and a learning rate of 0.001. The loss function was set as 'binary cross entropy' as the problem was of a binary classification. The optimizer 'RMS Prop' allowed the model to converge faster. The momentum value for optimizer was set as 0.9.

During the training process, even though initial loss values were high, the loss function started converging after 83 epochs and by 150 epochs, loss values for training and validation were 0.4 and 0.8, respectively (Fig. 14a). This also underlines the fact that there was no incidence of overfitting during the training process.

A similar pattern was observed in the accuracy plot during the training and validation process (Fig. 14b). The accuracy showed steady improvement in early epochs followed by a significant improvement after 80 epochs. The accuracy was comparatively steady after 110 epochs and the final values of accuracies for training and validation at the end of 150 epochs were 98.39 and 97.42 per cent, respectively. Further, the model was tested on the unseen 10 per cent of data and the results are recorded as a confusion matrix (Fig. 15).

The model has shown excellent prediction accuracy with testing data. Out of all 128 images on the testing dataset, the model has rightly identified classes of 125 images with an accuracy of 97.6 per cent and an F1-score of 97.78 per cent. Out of all 67 citrus fruits that belong to the "accept" class, the model could rightly identify 66 citrus fruits as the "accept" class and only 1 orange as the "reject" class. Similarly, the model could precisely classify 59 "reject" classes of citrus fruits out of 61 actual "reject" categories of citrus fruits. The precision and recall of the SortNet model were 97.0 and 98.5 per cent, respectively.

3.3. Performance comparison of SortNet with state-of-the-art CNN models

The performance of the SortNet model was compared with two state-of-the-art CNN models of varying complexity. AlexNet (Krizhevsky et al., 2012) and GoogLeNet (Szegedy et al., 2015) models were configured and evaluated in the same development environment. The architectural details of both AlexNet and GoogLeNet are given in Table 5. The developed SortNet model had an average testing time for a single fruit image on the raspberry pi 4 of 0.32 s; while the state-of-the-art models Alexnet, GoogLeNet required 0.43 and 0.56 s, respectively. SortNet required less training and testing; also it has a relatively less complex architecture. While comparing the accuracies of the models, GoogLeNet gave the best testing accuracy of 98.4 per cent during classification. Custom CNN delivered the second-best performance with an accuracy of 97.6 per cent, followed by AlexNet with 91.4 per cent, the lowest performance among all three classifiers (Table 6). SortNet and GoogLeNet both demonstrated an equivalent recall value of 98.5 per cent, a slight better when compared to AlexNet. However, the SortNet clearly outperforms the other state-of-the-art models (AlexNet and GoogLeNet) in terms of model simplicity, faster inference time and training time. This makes SortNet a more acceptable and savvier model for deploying in resource constraint edge devices like raspberry pi and to perform for accurate real-time classification of citrus fruits. Inference time was a key factor for the developed fruit grading system, as it would directly relate to the capacity of the machine. With SortNet, time and accuracy both remained uncompromised.

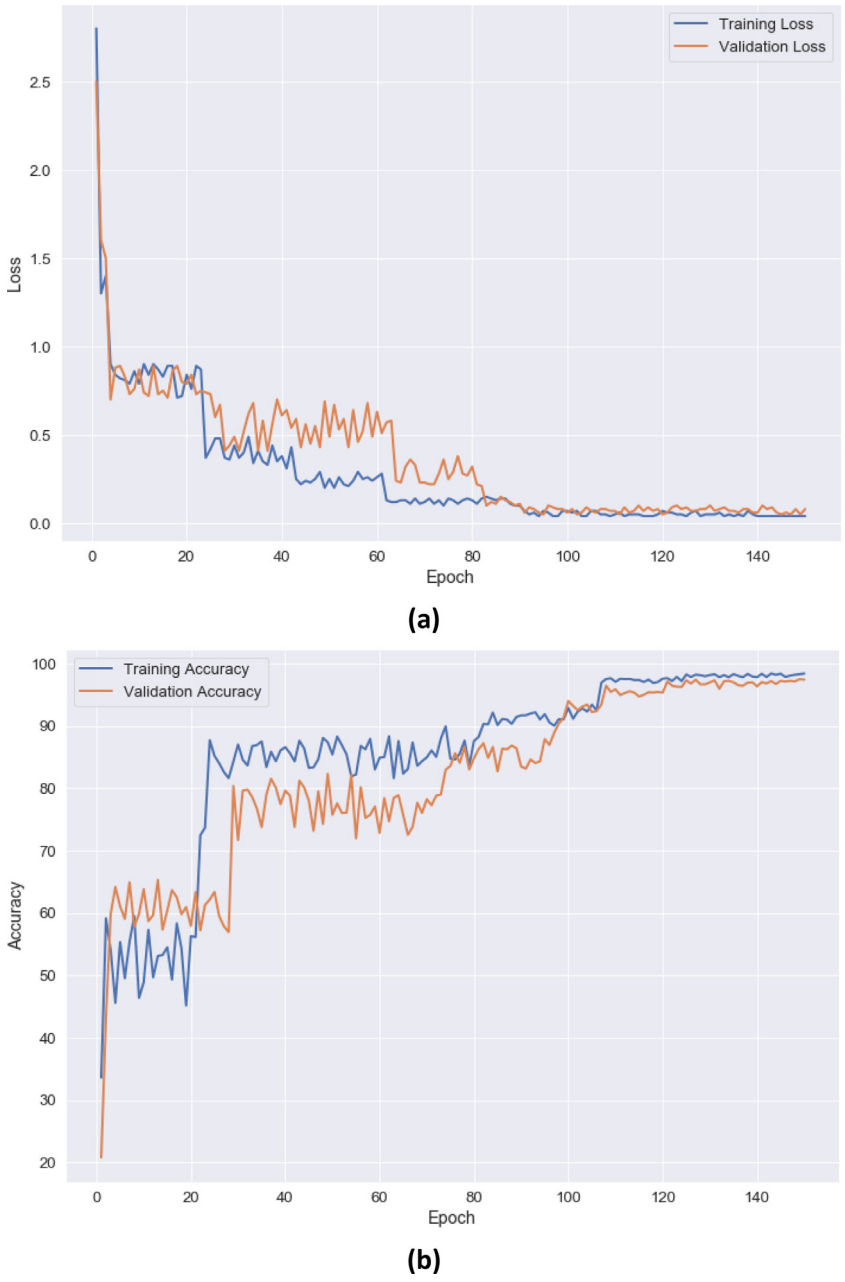


Fig. 14. Performance of SortNet model during training and validation (a) Loss graph (b) Accuracy graph.

Table 4
Confusion matrix for weight-based grading of citrus fruits.

Actual				
Predicted	Accuracy (%) = 91.3	Small (100 fruits)	Medium (100 fruits)	Large (100 fruits)
	Small	96 [0.96]	4	0
	Medium	3	88 [0.87]	10
	Large	0	8	90 [0.92]

Values in the square brackets indicate the "reliability".

Table 5
Architecture of state-of-the-art models and SortNet model.

DCNN models	Image input size	Parameters	Main layers	Train time (s)
Alexnet	227 × 227	6,23,78,344	5 conv, 3 FC layers	612
GoogLeNet	224 × 224	67,99,700	22 conv with inception layers	1342.8
SortNet	125 × 125	2,76,72,002	4 conv, 2FC layers	379.2

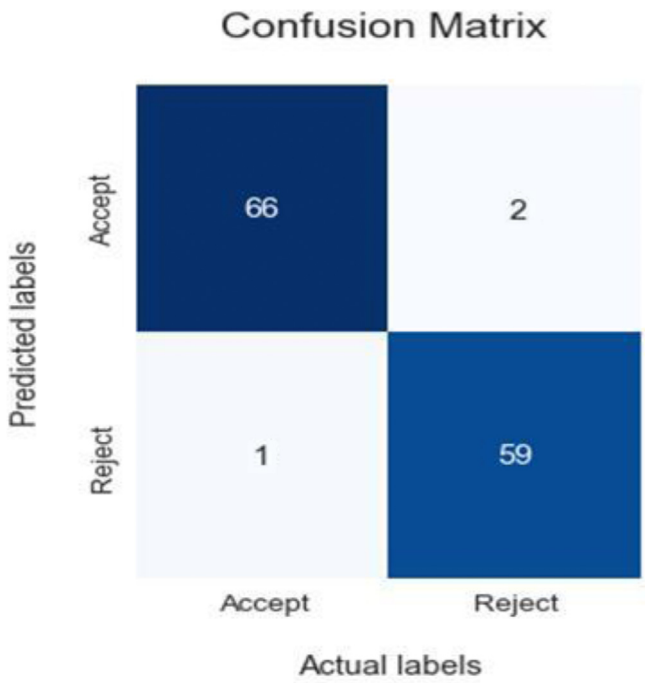


Fig. 15. Confusion matrix of SortNet model on testing dataset.

Table 6
Performance of state-of-the-art models and SortNet using testing dataset.

DCNN models	Accuracy (%)	Recall (%)	Precision (%)	Testing time (s)
Alexnet	91.4	88.0	95.1	11.7
GoogLeNet	98.4	98.5	98.5	13.1
SortNet	97.6	98.5	97.0	7.5

3.4. Ergonomic responses of the machine

A comparative analysis was conducted based on the responses while carrying out the tasks (washing, sorting and grading of citrus fruits) in traditional way (Fig. 16) with respect to that of the developed automatic machine.

3.4.1. Physiological responses

The observations (Fig. 17a) of physiological responses were understood in terms of work pulse (ΔHR, beat/min), oxygen consumption rate (OCR, L/min), energy consumption rate (ECR, kJ/min), VO_2 max (per cent) and cardiac cost (beat/min). The mean ΔHR was below 35 beats/min in case of traditional method as well as machine operation, this ΔHR value is within the acceptable limit for a sustainable work. Based on the values of OCR and ECR, the workload in traditional method qualifies for "moderate" to "heavy" across different components of the task, while the workload with machine was categorized as "moderate". The mean total cardiac cost for cleaning, sorting and grading of 100 kg citrus fruits with developed machine was found to be 1114 beats as against an average 3257 beats with traditional method. In other words, adoption of the developed machine can reduce cardiac cost by 82.9 per cent over the traditional method. It was observed that an average time taken by 12 subjects to work with 100 kg citrus fruits was about 58 min, while the machine took only 25 min; thus there was time saving of about 56 per cent.

3.4.2. Muscular responses

Maximum voluntary contraction (MVC) during the task for hand muscles, extensor radialis ulnaris (ERU)-left, extensor carpi ulnaris (ECU)-left, extensor radialis ulnaris (ERU)-right and extensor carpi ulnaris (ECU)-right responses of twelve subjects for the given task can be deciphered from Fig. 17b.



Fig. 16. Manual (with sensors strapped around the identified muscles to record the ergonomic parameters) traditional procedure for washing, vision-based sorting and grading of citrus fruits (a) Washing, (b) Collecting, (c) Manual sorting and grading.

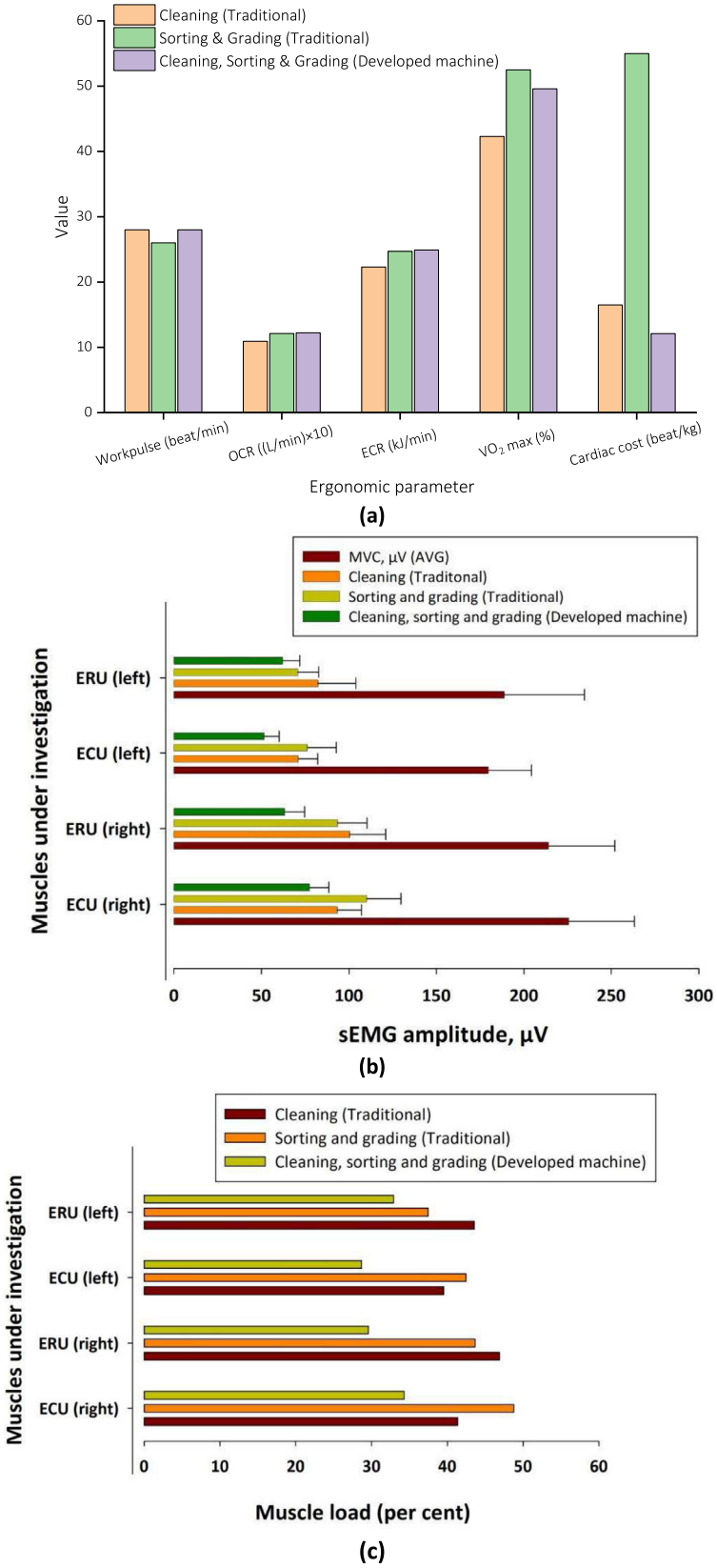


Fig. 17. A Comparative graphical representation of the various drudgery related indicators for cleaning, vision-based sorting and grading of citrus fruits with traditional and developed machine (a) Ergonomic parameters, (b) Muscle responses, (c) Muscle load. ERU (left)-Extensor radialis ulnaris (left hand), ECU (left)-Extensor carpi ulnaris (left hand), ERU (right)- Extensor radialis ulnaris (right hand); ECU (right)-Extensor carpi ulnaris (right hand), MVC-Maximum voluntary contraction.

The average MVC (*with* standard deviation) of hand muscles ERU (left), ECU (left), ERU (right) and ECU (right) were 188.79 (±45.89), 179.61 (±24.87), 214.12 (±37.94) and 225.64 (±37.61) μV, respectively. The responses of muscles under investigation were found to be lower with developed machine (51.49–77.42 μV) as compared to traditional method (70.74-110.07 μV).

The percentage load on muscles under investigation was observed to be higher for traditional method than it was for the developed machine (Fig. 17c). The percentage load on muscles during traditional method was recorded as high as 48.7 per cent which was higher than permissible muscle load of 35 per cent (Nag and Gite, 2020). However, muscles load ranged 28.67-34.31 per cent with the developed machine.

Thus, the hypotheses that introduction of "Automatic Fruit Grader" will decrease drudgery involved in washing, vision-based sorting and weight grading carried out manually stands correct and substantiated by the findings of this ergonomic study.

4. Conclusions

The aim of this study was to develop a light-weight DCNN model that can be easily deployed on an edge device like raspberry-pi to be used for real time vision-based sorting of citrus fruits in an optimally designed novel machine. The goals achieved in this study can be enumerated as, (i) a relatively less complex custom CNN model, "SortNet" was developed which could classify citrus fruits into predefined "accept" and "reject" with high accuracy and low response time. The precision and recall of the model were 97.0 and 98.5 per cent, respectively; (ii) "Automatic Fruit Grader" was able to carry out online real-time vision-based and three-category weight-based classification of citrus fruits with an accuracy of 97.0 and 91.3 per cent, respectively; (iii) the developed algorithm comprising a DCNN based architecture was robust enough to stand the test for on-farm postharvest activities.

It emerged that SortNet functioned far better than state-of-the-art pre-trained models that are relatively more complex and fail to provide the low response time required for online applications with low-cost cameras and controllers. The optimally designed and developed "Automatic Fruit Grader" achieved good accuracy and drastically reduced the drudgery (by more than 82 per cent) associated with similar manual operations. This machine can find widespread application for postharvest operation automation in pack houses.

CRediT authorship contribution statement

Subir Kumar Chakraborty: Conceptualization, Methodology, Formal analysis, Validation, Investigation, Writing – original draft, Writing – review & editing, Project administration. **Subeesh A.:** Software development, Validation, Investigation, Writing – review & editing. **Kumkum Dubey:** Design of electronic circuits, Testing of the machine, Validation, Writing – review & editing. **Dilip Jat:** Design of machine parameters, Optimization experiments and analysis, Validation, Writing – review & editing. **Narendra Singh Chandel:** Conceptualization, Formal analysis, Data curation, Writing – review & editing. **Rahul Potdar:** Ergonomics, Experiments, Data collection and analysis, Writing – review & editing. **N.R.N.V. Gowripathi Rao:** Dimensional analysis, Machine design, Data analysis, Writing – review & editing. **Deepak Kumar:** Design of electronic circuits, Testing of the machine, Validation, Data collection.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.engappai.2023.105826.

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